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AI-Driven Sustainable Competitive Advantage in Tourism and Hospitality: Mediating Roles of Digital Culture and Skills

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Abstract

This study explored how AI affects the sustainability of competitive advantage in the tourism and hospitality sector, with a particular focus on the mediating roles of digital culture and digital skills in the lens of the Technology Acceptance Model (TAM). Data were collected via a structured questionnaire distributed to a purposive sample of 488 managers and supervisors working in five-star hotels, travel agencies, and DMCs across Saudi Arabia. The findings revealed that AI has a significant direct effect on sustainable competitive advantage and also exerts strong positive effects on both digital culture and digital skills. In turn, both of these internal enablers significantly contribute to sustaining a competitive advantage. Mediation analysis further showed that both digital culture and digital skills partially mediate the relationship between AI and sustainable competitiveness. The study addresses a notable gap in tourism research by providing localized evidence from a market undergoing rapid transformation under Vision 2030, and, taken together, extends TAM to an organizational lens by demonstrating AI's role in shaping culture and skills that underpin a durable advantage while pointing to actionable priorities—targeting high-value AI use cases, conducting capability audits, institutionalizing continuous learning through visible leadership and role-based upskilling, and embedding culture- and skills-oriented KPIs within AI governance.

Keywords: artificial intelligence; digital culture; digital skills; sustainable competitive advantage; tourism and hospitality

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1. Introduction

In the era of digital transformation, achieving and sustaining competitive advantage in tourism and hospitality require more than the simple adoption of new technologies. This calls for a broader strategic shift that integrates advanced tools such as AI with organizational culture and workforce capabilities [1]. The strength of AI lies in its ability to create value across several dimensions. In customer service, for example, AI supports hyper-personalized guest experiences through recommendation engines and sentiment analysis, which in turn enhance satisfaction and loyalty [2]. In operations, AI-driven automation helps reduce costs and optimize processes, including dynamic pricing, staff scheduling, and energy management in hotels [3]. At the strategic level, AI enables data-driven decision-making, giving leaders the ability to anticipate market trends and respond effectively to shifts in consumer behavior. When these capabilities are aligned with business

objectives, they create sustainable differentiation in an increasingly competitive and fast-changing sector [4]. Recent studies have shown that AI is being applied in multiple areas of the tourism and hospitality sector, from chatbots that streamline reservations and provide real-time customer support, to predictive analytics that help forecast demand and optimize pricing strategies. In the hotel sector, AI-driven energy management systems contribute to sustainability goals by reducing utility consumption, while smart service robots are increasingly being used to enhance guest interactions and improve service efficiency. In airlines and travel agencies, AI supports personalized marketing campaigns and improves customer relationship management by analyzing vast datasets of traveler preferences. Collectively, these applications indicate that AI is no longer an experimental tool but a mainstream driver of competitiveness in global tourism markets [1,3,4].

However, the long-term effectiveness of AI depends on the presence of a strong digital culture and a workforce equipped with robust digital skills [5]. Digital culture reflects an organization's collective mindset—its openness to experimentation, data-driven innovation, and agility in adopting new tools [6]. AI itself can influence this culture by encouraging employees to rely more on predictive insights and by fostering cross-functional collaboration during implementation [7]. At the same time, AI raises the demand for digital skills, which include both technical abilities such as data interpretation, AI literacy, and digital communication, and soft skills such as adaptability and digital leadership. These competencies are increasingly essential for employees at all levels to engage with AI in meaningful ways and to drive organizational transformation [8,9]. The relationship is reciprocal: a strong digital culture and a digitally skilled workforce not only support the successful adoption of AI but also shape its broader impact on organizational outcomes. When employees combine relevant skills with a cultural alignment toward innovation, organizations are better positioned to generate sustained innovation, enhance service quality, and adapt to disruptions—all of which are key elements of long-term competitive advantage [10]. In line with these developments, recent research emphasizes that a digitally skilled workforce and a culture that embraces innovation are critical in determining whether AI adoption delivers meaningful outcomes [2,5,7]. For instance, organizations with strong digital leadership and openness to experimentation are more likely to embed AI into strategic processes, rather than limiting it to isolated pilot projects. Similarly, employees with higher levels of AI literacy and data-driven decision-making skills are better equipped to extract actionable insights from AI systems, thus amplifying their strategic value. Without these cultural and human foundations, even advanced AI solutions risk underutilization or resistance, leading to wasted investment and minimal long-term impact [9].

This interplay aligns closely with the TAM, which posits that the perceived usefulness and perceived ease of use of technology influence users' attitudes and intentions to adopt it [11]. Within the context of AI in tourism and hospitality, TAM helps explain how employees and managers accept or resist AI tools based on their understanding, digital competence, and cultural readiness [12]. A digital culture that promotes openness to innovation and continuous learning increases the perceived ease of use, while visible benefits of AI in customer satisfaction or efficiency boost perceived usefulness [13]. In this study, AI represents the technological stimulus that shapes both perceived usefulness and ease of use, while digital culture and digital skills function as organizational enablers that translate these perceptions into actual behavioral and strategic outcomes. By influencing how employees evaluate and engage with AI, these two variables operationalize the social and cognitive pathways through which TAM unfolds at the organizational level. Although TAM was originally designed to capture individual-level adoption, its adaptation here acknowledges that acceptance is not only a personal cognitive judgment but also an organizational process shaped by shared values and workforce capabilities. By embedding digital

culture and digital skills into the model, the study demonstrates how TAM constructs can be extended to explain collective readiness for AI and its translation into sustainable performance outcomes. Thus, TAM provides a theoretical bridge linking the technological (AI), cultural (digital mindset), and human (skills) elements that collectively determine the sustainability of competitive advantage. As well, positioning sustainable competitive advantage as the ultimate outcome extends TAM beyond individual acceptance, showing how perceptions of usefulness and ease of use evolve into long-term organizational capabilities when mediated by culture and skills. Moreover, by integrating organizational-level enablers into TAM, this study goes beyond prior applications of the model and shows how micro-level perceptions evolve into meso- and macro-level strategic outcomes—offering a novel theoretical extension that connects technology adoption frameworks with strategic management theories.

Despite a growing body of research on AI adoption in service industries (e.g., [1,3,9]), there remains limited understanding of how AI contributes to the sustainability of competitive advantage when applied within organizational contexts—particularly in the tourism and hospitality sectors of emerging markets such as Saudi Arabia. Much of the existing literature has concentrated on short-term performance outcomes (e.g., [12,14]), while giving less attention to the internal enablers that determine AI's long-term strategic value. A relatively underexplored area in the literature concerns the mediating role of digital culture and digital skills in turning AI capabilities into sustainable competitive advantage. Much of the existing work discusses these elements separately—AI is often studied from a technical or consumer angle, while digital culture and skills are framed as general HR or organizational issues—leaving few attempts to examine them together in one framework. This separation limits the development of integrated strategies for digital maturity in tourism and hospitality organizations, particularly in contexts where digital capability is still emerging. By explicitly linking these dimensions, the present study offers a theoretical integration that has been largely absent in prior research.

TAM has been widely applied to understand technology adoption, but the focus has remained mostly on customer-facing technologies such as booking platforms and mobile applications. Far fewer studies explore how managers and employees engage with AI systems inside service organizations [15,16]. This creates room to extend TAM by linking its central constructs—perceived usefulness and perceived ease of use—to organizational factors such as digital culture and digital skills, and to show how these shape both AI adoption and longer-term competitive outcomes. In addition, our conceptual framing draws on the Nature-Inspired System perspective, which emphasizes adaptive and evolutionary mechanisms observed in natural phenomena, such as self-organization in biological systems, swarm intelligence, and resilience under changing environments. The key premise of this perspective is that organizations, much like natural systems, survive and thrive by continuously learning, evolving, and adjusting to external stimuli [9,16]. Applied to the tourism and hospitality context, this suggests that firms adopting AI are not static but rather adaptive entities that reshape their digital culture and workforce skills in response to technological disruptions [9,16]. By integrating TAM with a nature-inspired view, the study positions AI-driven adoption as both a matter of perceived usefulness/ease and as part of a broader adaptive cycle that underpins sustainable competitive advantage. This dual-theory approach not only bridges adoption and adaptation logics but also introduces a hybrid framework that enriches both TAM and systems-based theories of organizational resilience.

In Saudi Arabia, these questions take on particular importance. Tourism has been positioned as a key pillar of Vision 2030, and major investment in AI and digital infrastructure is reshaping the sector [17,18]. The rapid expansion of hospitality projects alongside a national digital transformation agenda makes this a timely setting to study how technology, culture,

and skills interact to create sustained competitive advantage. Yet, empirical studies from this context remain limited, highlighting the need for localized, evidence-based research.

In response to these gaps, the study pursues three interrelated objectives. First, it examines the effect of AI on sustainable competitive advantage, digital culture, and digital skills in tourism and hospitality organizations. Second, it explores the influence of digital culture and digital skills on sustaining competitive advantage. Third, it investigates the mediating roles of these two enablers—digital culture and digital skills—in the relationship between AI and sustainable competitive advantage. This research addresses important gaps in understanding how AI can contribute to long-term competitiveness in Saudi Arabia's tourism and hospitality sector. By focusing on the mediating roles of digital culture and digital skills, it extends the TAM to the organizational level, emphasizing internal adoption of AI. Overall, the theoretical contribution of this study lies in: (1) extending TAM from an individual adoption model to an organizational-level framework; (2) integrating digital culture and digital skills as mediators that connect AI adoption with sustainable competitive advantage; and (3) advancing a dual-theory approach that combines TAM with the Nature-Inspired System perspective to explain both adoption and adaptation mechanisms. Beyond addressing the empirical gaps, this study also advances theory by offering a novel integrative framework that unifies AI adoption, digital culture, and digital skills under a dual-theory lens. While previous research has often treated these elements in isolation, our model moves the debate from fragmented insights toward a holistic explanation of how technological adoption translates into enduring strategic capabilities. Importantly, this theoretical advancement extends beyond the Saudi context, contributing to the broader international discourse on AI-driven sustainable competitive advantage in service industries. The study is expected to deliver a validated model linking AI, digital enablers, and strategic outcomes; contribute theoretical insights; and provide practical guidance for enhancing digital readiness in alignment with Saudi Vision 2030.

The remainder of the paper is organized as follows: Section 2 reviews the literature and develops the hypotheses; Section 3 describes the sample and data collection, measurement instruments, and common method bias checks; Section 4 presents the empirical results (measurement and structural models); Section 5 discusses the findings; Section 6 outlines theoretical implications; Section 7 details practical implications; and Section 8 concludes with limitations and avenues for future research.

2. Literature Review and Hypotheses Development

2.1. The Effect of AI on Sustainable Competitive Advantage

Sustainable competitive advantage is the ability of a business to preserve its edge in the market over the long term by relying on distinctive resources and capabilities that are difficult for competitors to replicate. It goes beyond short-lived advantages and focuses on maintaining value through innovation, adaptability, and effective use of organizational strengths. By aligning strategies with economic, social, and environmental considerations, firms can secure lasting success while building trust with their stakeholders [19]. Sustaining long-term strategic advantage increasingly depends on the ability to turn raw data into actionable insights that enhance guest experiences, optimize resources, and support predictive decision-making in tourism and hospitality [16]. When AI is embedded into service and operational systems as adaptive and learning infrastructures, it generates outcomes that are difficult for competitors to imitate [20]. These outcomes evolve over time, becoming more personalized, efficient, and aligned with organizational goals, thereby creating a competitive position that is strengthened through continuous feedback and learning [9]. The TAM provides a useful lens for understanding how AI contributes to this sustainability [21]. Perceived usefulness—such as improvements in task performance

and decision-making speed—directly influences employees' and managers' willingness to adopt AI [22]. Similarly, when AI systems are perceived as easy to learn and integrate into workflows, adoption is more likely to be successful and widespread [23]. AI has been shown to positively influence sustainable competitive advantage [24], act as a strategic enabler of long-term competitiveness [25], and serve as a key resource that helps organizations maintain their position in rapidly changing markets [26]. Accordingly, the following hypothesis can be proposed:

H1. *AI has a positive effect on sustainable competitive advantage.*

2.2. The Effect of AI on Digital Culture

Digital Culture refers to the shared values, practices, and ways of thinking that emerge from the use of digital technologies in daily life and work. It shapes how individuals and organizations communicate, collaborate, and innovate in a technology-driven environment [27]. When intelligent technologies become embedded in daily tasks and decision-making, they gradually reshape the behavioral norms and shared assumptions of organizations. AI-driven tools such as real-time service optimization, facial recognition check-ins, and predictive guest preference models do more than enhance efficiency—they influence collaboration, approaches to risk, and attitudes toward innovation in tourism and hospitality [16]. These systems encourage employees to rely less on intuition, consult data-based evidence, and adopt a more experimental mindset. Over time, repeated behaviors of this kind solidify into organizational habits and shared beliefs, creating a digitally adaptive environment where learning, openness to technology, and agility form the cultural baseline [9]. The TAM explains this process by emphasizing that individuals' perceptions of usefulness and ease of use determine whether technologies are adopted and embedded [28]. Yet these perceptions rarely remain individual; instead, they often cascade into collective patterns that reinforce culture [29]. Empirical evidence supports this dynamic: [14] found that AI adoption strengthens digital culture, while [30] showed that integrating AI technologies fosters a digitally oriented mindset. Similarly, Ref. [31] concluded that AI accelerates the internalization of digital values and enhances organizational alignment with digital transformation, and [32] argued that AI acts as a transformative agent that reinforces digital culture. Therefore, the following hypothesis can be proposed:

H2. *AI has a positive effect on digital culture.*

2.3. The Effect of Digital Culture on Sustainable Competitive Advantage

Organizations that foster digital awareness, openness to innovation, and data-driven thinking are better positioned to remain competitive in rapidly evolving markets [33]. In tourism and hospitality, where customer expectations change quickly and operations are influenced by seasonality and global dynamics, a culture that encourages digital experimentation becomes a strategic advantage. Such a culture promotes the proactive use of technology, faster adaptation to market intelligence, and continuous refinement of services based on digital insights and customer feedback [34]. The flexibility and responsiveness that follow allow firms to introduce service innovations more rapidly than competitors, improve internal efficiency, and deliver personalized guest experiences—all of which strengthen their ability to sustain a competitive edge [35]. This cultural influence is further reinforced by how individuals perceive and respond to technology. According to TAM, technology acceptance depends on perceptions of usefulness and ease of use, and in digitally oriented organizations these perceptions are positively amplified [36]. Empirical evidence confirms this link: [37] showed that digital culture significantly enhances long-term competitive advantage; [38] found that it supports the development of sustainable competitiveness; [39]

highlighted its role in strengthening strategic flexibility and innovation capacity; and [40] together with [41] emphasized its strategic importance in reinforcing sustainable competitive advantage. Based on this dynamic, the following hypothesis can be proposed:

H3. *Digital culture has a positive effect on sustainable competitive advantage.*

2.4. The Mediating Role of Digital Culture

The value of AI in tourism and hospitality is not derived from the technology alone but from the organizational setting in which it is applied [36]. Tools such as booking engines, service automation, or predictive maintenance generate far greater benefits when introduced into cultures that already prize digital curiosity, adaptability, and openness to experimentation [42]. In these contexts, culture becomes the medium through which technical capabilities are translated into agility, collaboration, and knowledge sharing [9]. Staff are more inclined to interpret data creatively, use AI insights to enrich guest experiences, and cooperate across units through digital platforms. The result is not only improved efficiency but also a greater capacity for strategic renewal [42]. TAM helps explain why this occurs. Perceptions of usefulness and ease of use are rarely shaped by individuals alone; they are influenced by peers, leaders, and the wider organizational ethos around technology [43]. When cultural norms encourage digital engagement, employees are more likely to see AI as beneficial and less likely to perceive it as disruptive, which accelerates its integration into decision-making and service design [44]. In this way, digital culture acts as a filter: it determines whether AI becomes a driver of transformation or remains underused [42]. In light of this reasoning, the following hypothesis can be proposed:

H4. *Digital culture mediates the relationship between AI and sustainable competitive advantage.*

2.5. The Effect of AI on Digital Skills

Digital Skills are the practical abilities and competencies needed to use digital tools and technologies effectively [45,46]. They include basic skills such as information search and online communication, as well as advanced abilities like data analysis, digital content creation, and cybersecurity awareness [47,48]. As intelligent technologies become embedded in tourism and hospitality operations, they inevitably reshape the skills required from the workforce [49]. Employees are no longer confined to standardized procedures; they are increasingly expected to interpret outputs from AI systems, make real-time decisions, and interact with guests through digitally mediated platforms [27]. Continuous exposure to these tools fosters experiential learning, gradually building digital fluency and technical confidence [50]. In this sense, AI functions not only as a driver of operational change but also as a catalyst for upskilling and the ongoing development of workforce capabilities [51]. The TAM helps explain this process. Its constructs—perceived usefulness and perceived ease of use—act as motivational levers that shape employees' willingness to engage with AI and sustain learning behaviors [52,53]. Evidence from recent studies further illustrates this dynamic: [54] shows that AI integration enhances employees' digital skills, while [55] emphasizes its role in accelerating skill advancement. Reference [56] demonstrates that embedding AI in organizational processes supports systematic capability building, and [57] highlights its substantial contribution to strengthening digital proficiency across the workforce. Based on this logic, the following hypothesis can be proposed:

H5. *AI has a positive effect on digital skills.*

2.6. The Effect of Digital Skills on Sustainable Competitive Advantage

The ability of organizations to sustain a competitive position increasingly depends on the digital readiness of their workforce [58]. In tourism and hospitality, where services are both time-sensitive and highly personalized, employees with strong digital skills can leverage technology not only to improve operational efficiency but also to enhance customer engagement [59]. A digitally skilled workforce also contributes to greater organizational agility, enabling faster implementation of innovations, proactive problem-solving, and scalable personalization without compromising quality [60]. TAM helps explain this dynamic, suggesting that perceptions of usefulness and ease of use drive technology adoption, and that repeated, meaningful interaction with digital systems gradually fosters skill acquisition [61]. In this sense, digital skills emerge not only from formal training but also from supportive environments that encourage experimentation, reduce fear of failure, and reinforce the practical value of technology. Over time, these competencies evolve into strategic capabilities that underpin long-term success [19]. Empirical studies provide consistent evidence of this relationship. Reference [62] reports that digital skills positively influence sustainable competitive advantage. Reference [63] found that such skills are particularly critical in SMEs facing technological uncertainty. Reference [19] emphasizes that digital talent and knowledge workers are pivotal for sustaining competitiveness in tourism and hospitality. Similarly, Ref. [64] argues that strengthening workforce digital competencies is essential for achieving long-term competitive advantage. Therefore, the following hypothesis can be proposed:

H6. *Digital skills have a positive effect on sustainable competitive advantage.*

2.7. The Mediating Role of Digital Skills

Integrating AI into tourism and hospitality operations creates both opportunities and challenges [65]. Technologies such as personalization engines, predictive analytics, and service automation hold significant potential, but their impact depends largely on how effectively employees can work with them [9]. The presence of AI often accelerates shifts in workforce expectations, requiring staff to develop new competencies, adapt to evolving systems, and take a more proactive role in problem-solving and decision-making [66]. As digital skills expand alongside AI adoption, they serve as a lever that converts technological potential into measurable performance. In this way, organizations are not only deploying tools but also building the capacity to innovate, adapt, and sustain competitiveness in technology-driven markets [9]. The TAM highlights the role of user perceptions in shaping adoption outcomes [67]. When employees view AI systems as useful and manageable, they are more likely to engage with them consistently and confidently, creating opportunities for informal learning and skill development [68]. Over time, these skills enable employees to derive greater value from AI, apply systems in creative ways, and strengthen the organization's agility [60]. Without such evolving capabilities, AI risks being underutilized and disconnected from broader strategic goals. Digital skills therefore operate as a behavioral bridge that links technological potential with long-term competitive advantage [19]. Accordingly, the following hypothesis can be proposed:

H7. *Digital skills mediate the relationship between AI and sustainable competitive advantage.*

The following proposed research model is visually summarized in Figure 1. The figure illustrates the hypothesized relationships (H1–H7) between AI, digital culture, digital skills, and sustainable competitive advantage.

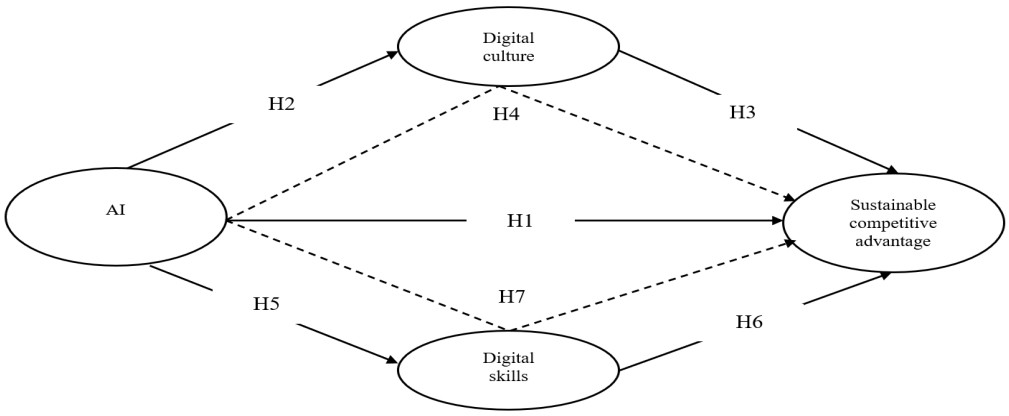

Figure 1. Conceptual Research Model.

3. Methods

3.1. Sample and Data Collection

Saudi Arabia's tourism and hospitality sector has accelerated under Vision 2030, with the Ministry of Tourism reporting ~116 million domestic and inbound tourist trips in 2024—a new national record—supported by rising international visitor spending and a broadened product mix across premium hotels, licensed travel agencies, and tour operators (including destination management companies—DMCs). Five-star hotels anchor the upper tier of supply and service quality, while travel agencies and DMCs structure and package experiences, operate reservations and ground services, and connect visitors to destinations across the Kingdom under the Ministry's licensing framework. At the macro level, UN Tourism (UNWTO) recognized Saudi Arabia's surpassing of 100 million tourist visits as a milestone in 2023, underscoring momentum that continued through 2024; Ministry releases also note double-digit growth in inbound visitor spending [19,69–73]. To ensure contextual relevance, we surveyed decision-makers and supervisors in Saudi Arabia's five-star hotels, licensed travel agencies, and DMCs—the parts of the sector that make day-to-day operational and digital choices. This frame fits the study's focus on internal AI adoption for three reasons. First, five-star hotels operate under detailed national classification standards that embed digital service requirements (e.g., automated guest-register software, free lobby Wi-Fi, mobile/digital check-in and express checkout), indicating advanced technology use at the premium end of supply. Second, travel agencies and tour-operations firms function within the Ministry of Tourism's licensing regime—which specifies activity categories and bank-guarantee thresholds—signaling formalized, digitized processes and higher service complexity in packaging, reservations, and ground services [19,69–71,73,74]. Third, the subsectors we targeted are central to the National Tourism Strategy under Vision 2030, which positions tourism—and its digital enablement—as a pillar of economic transformation.

Although the research population was defined by the number of licensed institutions (five-star hotels, travel agencies, and DMCs) (Riyadh: 34; Jeddah: 20; Makkah: 21; Madinah: 20; Eastern Province: 16), 206 licensed travel agencies (Riyadh: 68; Jeddah: 48; Makkah: 34; Madinah: 17; Eastern Province: 39), and 82 licensed DMCs (Riyadh: 25; Jeddah: 26; Makkah: 13; Madinah: 15; Eastern Province: 3) reported by the [74], the actual sampling unit was managers and supervisors with at least one year of tenure within those organizations. Purposive sampling was employed, with strata used only to ensure coverage across subsectors and regions. Exact counts of eligible managers and supervisors per stratum were not available, and the study therefore acknowledges limitations in representativeness. The final valid sample was proportionally distributed to mirror the underlying population: 196 respondents from five-star hotels, 178 from travel agencies, and 114 from

DMCs. Appendix A provides a full cross-tabulation of both the population figures and the corresponding sample distribution by subsector and region.

The sample size was primarily determined based on the requirements of Structural Equation Modeling (SEM), which was employed as the main data analysis technique. SEM is a powerful statistical method that requires an adequate sample size to ensure estimation accuracy and model stability. According to well-established guidelines [75], a minimum of 10 observations per item is generally recommended for robust SEM analysis. As the final questionnaire included 34 measurement items, the required minimum sample size was at least 340 valid responses to ensure analytical rigor and statistical power. Participation was strictly voluntary. An information sheet preceded the questionnaire outlining the study purpose, expected time, minimal risks, confidentiality safeguards, and the right to withdraw at any time without penalty; informed consent was recorded before respondents proceeded. No names, emails, or employer-identifying details were collected, and responses were reported only in aggregate. Although participants were formally allowed to skip any question to reduce potential discomfort and protect privacy, in practice all respondents completed every item, yielding 0% missing data and ensuring that eligibility criteria were verified for the entire sample. No incentives were offered beyond an optional summary of findings. The study adhered to recognized standards for research with human participants and to journal guidance on ethics and data protection.

Data collection ran for two months (April–May 2025). Distribution was managed by four trained assistant researchers who coordinated with designated gatekeepers (HR and operations managers) in participating five-star hotels, travel agencies, and DMCs across major Saudi cities. The gatekeepers circulated a paper-based version of the questionnaire to eligible managers and supervisors, and completed forms were collected by the assistant researchers in sealed envelopes to ensure confidentiality. Neither the authors nor the assistants held or used detailed personal contact information for the 680 potential respondents. In total, 680 questionnaires were issued through these organizational channels, and 488 complete, valid responses were returned and retained for analysis, yielding a valid response rate of 71.8%.

The majority of participants were male (74%), while females constituted only 26% of the sample. In terms of age distribution, the largest group (43%) was aged between 40 and less than 50 years, followed by 34.2% between 30 and less than 40 years. Regarding educational attainment, an overwhelming majority (88.9%) held a bachelor's degree, while only 4.5% had postgraduate qualifications. Most respondents (77.7%) were supervisors, compared to 22.3% who were managers. Finally, the data indicate that nearly half of the participants (49.6%) had 10 to less than 15 years of professional experience, with only 3.5% having less than five years.

3.2. Measures

The measurement instrument for this study was designed based on previously validated scales to ensure both content relevance and construct validity. The questionnaire covered the study's four key constructs: AI, digital organizational culture, digital skills, and sustainable competitive advantage. All items were rated using a 5-point Likert scale ranging from (1) "strongly disagree" to (5) "strongly agree". The construct of AI adoption was measured using a 10-item scale adapted from [9]. Sample items included statements such as: "AI systems are integrated with existing business processes and workflows". Digital culture (DC) was assessed using a 4-item scale adapted from [76]. A sample item from this construct is: "The teams collaborate functionally in the initiatives for the innovation and digital transformation." The measurement of digital skills (DS) was conducted using a 15-item scale adapted from the works of [77,78]. Example items include: "The company

has high-performing employees who can complete routine tasks". Lastly, sustainable competitive advantage was measured using a 4-item scale developed by [79]. Sample items include: "This hotel/travel agency offers comparatively lower prices than competitors". In addition, the questionnaire included a section for collecting demographic data.

3.3. Common Method Bias

To minimize the potential impact of common method bias (CMB)—a concern in self-reported, cross-sectional survey designs—several procedural and statistical remedies were employed. Procedurally, anonymity and confidentiality were assured to all respondents to reduce social desirability and evaluation apprehension. In addition, the questionnaire incorporated mixed item wording and varied scale formats to limit response patterns and psychological priming [75]. Statistically, Harman's single-factor test was conducted, and the results indicated that no single factor accounted for the majority of the variance, suggesting that CMB was not a serious threat. Furthermore, full collinearity assessment was performed using variance inflation factor (VIF) values, all of which were below the recommended threshold of 3.3, confirming the absence of substantial common method bias [80,81].

4. Results

4.1. Measurement Model

To ensure the robustness of the structural model and the validity of its underlying constructs, the measurement model was rigorously evaluated using the full sample. The assessment focused on internal consistency reliability, convergent validity, and factor structure through confirmatory factor analysis. The overall model demonstrated a strong fit with the observed data, as evidenced by multiple fit indices: CMIN/DF was 1.967, which is well below the recommended threshold of 3.0; other indices further confirmed model adequacy, including GFI = 0.960, CFI = 0.948, IFI = 0.944, NFI = 0.951, TLI = 0.954, and RMSEA = 0.017—all of which fall within acceptable to excellent ranges, indicating a good model-data fit [75].

As shown in Table 1, all four constructs exceeded the recommended reliability thresholds. AI reported a Cronbach's alpha of 0.876, composite reliability (CR) of 0.967, and average variance extracted (AVE) of 0.743. These values reflect a high degree of internal consistency and convergent validity, with individual item loadings ranging from 0.808 to 0.903—all exceeding the 0.70 standard. Similarly, digital culture showed satisfactory reliability (α = 0.791, CR = 0.904), and its AVE of 0.701 supports adequate variance explanation by the latent construct. All item loadings were above 0.79, confirming strong construct representation as [75] confirmed.

Table 1. Measurement model.

Constructs	Factor Loading	Cronbach's Alpha	CR	AVE
Artificial intelligence		0.876	0.967	0.743
AI1	0.887			
AI2	0.903			
AI3	0.883			
AI4	0.891			
AI5	0.811			
AI6	0.856			
AI7	0.881			
AI8	0.862			
AI9	0.832			
AI10	0.808			
Digital culture		0.791	0.904	0.701
DC1	0.799			
DC2	0.841			
DC3	0.829			

Table 1. *Cont.*

Constructs	Factor Loading	Cronbach's Alpha	CR	AVE
DC4	0.878			
Digital skills		0.883	0.976	0.730
DS1	0.834			
DS2	0.871			
DS3	0.882			
DS4	0.809			
DS5	0.811			
DS6	0.845			
DS7	0.890			
DS8	0.826			
DS9	0.877			
DS10	0.817			
DS11	0.854			
DS12	0.810			
DS13	0.862			
DS14	0.922			
DS15	0.893			
Sustainable competitive advantage		0.851	0.921	0.746
SCA1	0.889			
SCA2	0.861			
SCA3	0.837			
SCA4	0.866			

Source: Developed by authors.

The construct of digital skills exhibited exceptional psychometric properties, with Cronbach's alpha reaching 0.883 and CR at 0.976, well above the 0.90 threshold, suggesting excellent reliability. The AVE was 0.730, and all 15 items demonstrated strong loadings between 0.809 and 0.922, indicating that this construct is measured comprehensively and reliably. Finally, sustainable competitive advantage also met the validity criteria, with α = 0.851, CR = 0.921, and AVE = 0.746. The four corresponding items displayed high loadings, ranging from 0.837 to 0.889, confirming the construct's conceptual clarity and empirical strength [75].

To further establish the distinctiveness of the study's constructs, discriminant validity was assessed using both the Fornell–Larcker criterion (Table 2) and the Heterotrait–Monotrait ratio (HTMT, Table 3). According to the Fornell–Larcker criterion, each construct's square root of AVE (shown on the diagonal) exceeds its highest correlation with any other construct: for example, AI's $\sqrt{\text{AVE}}$ (0.862) is greater than its correlation with digital culture (0.438), digital skills (0.449), and sustainable competitive advantage (0.601). Similarly, digital culture (0.837), digital skills (0.854), and sustainable competitive advantage (0.863) all surpass their inter-construct correlations, confirming that each latent variable shares more variance with its own indicators than with other constructs [82].

Table 2. Discriminant validity.

	A1	Digital Culture	Digital Skills	Sustainable Competitive Advantage
AI	**0.862**			
Digital culture	0.438	**0.837**		
Digital skills	0.449	0.388	**0.854**	
Sustainable competitive advantage	0.601	0.571	0.502	**0.863**

Source: Developed by authors.

Table 3. HTMT for validity.

	A1	Digital Culture	Digital Skills	Sustainable Competitive Advantage
AI				
Digital culture	0.398			
Digital skills	0.444	0.519		
Sustainable competitive advantage	0.487	0.554	0.421	

Source: Developed by authors.

Complementing this, the HTMT values in Table 3 all fall well below the conservative threshold of 0.85 (ranging from 0.398 to 0.554), providing additional evidence that the constructs are empirically distinct. In particular, the highest HTMT ratio—between digital culture and sustainable competitive advantage—is 0.554, comfortably under the cutoff. Together, these results demonstrate robust discriminant validity, ensuring that AI, digital culture, digital skills, and sustainable competitive advantage are each measured as separate, non-overlapping dimensions within the model [75].

4.2. Structure Model

The results of the structural model, as presented in Table 4, offer strong empirical support for all proposed hypotheses, thereby confirming the theoretical framework of the study. Starting with the direct effects, H1 was confirmed, as AI demonstrated a statistically significant and positive influence on sustainable competitive advantage (β = 0.527, CR = 6.934, $p < 0.001$). Furthermore, H2 and H5 were also supported, with AI showing significant effects on both digital culture (β = 0.458, CR = 5.725, $p < 0.001$) and digital skills (β = 0.509, CR = 6.207, $p < 0.001$), respectively.

Table 4. Hypotheses tests.

Path	*β*	s.e	CR	*p*-Value	Results
H1: AI → Sustainable competitive advantage	0.527	0.076	6.934	>0.001	Accepted
H2: AI → Digital culture	0.458	0.080	5.725	<0.001	Accepted
H3: Digital culture → Sustainable competitive advantage	0.421	0.069	6.101	<0.001	Accepted
H5: AI → Digital skills	0.509	0.082	6.207	<0.001	Accepted
H6: Digital skills → Sustainable competitive advantage	0.441	0.075	5.880	<0.001	Accepted
Mediating effect					
H4: A1 → Digital culture → Sustainable competitive advantage	0.193	0.050	3.860	<0.001	Accepted
H7: A1 → Digital skills → Sustainable competitive advantage	0.224	0.047	4.766	<0.001	Accepted

Source: Developed by authors.

In addition, the results confirmed H3 and H6, as both digital culture (β = 0.421, CR = 6.101, $p < 0.001$) and digital skills (β = 0.441, CR = 5.880, $p < 0.001$) were found to significantly enhance sustainable competitive advantage. The mediating hypotheses, H4 and H7, were also empirically validated. The indirect effect of AI on sustainable competitive advantage via digital culture (β = 0.193, CR = 3.860, $p < 0.001$) and via digital skills (β = 0.224, CR = 4.766, $p < 0.001$) were both statistically significant. Importantly, because the direct path from AI to sustainable competitive advantage (H1) remained significant even in the presence of both mediators, the mediation is classified as partial, not full. This implies that while digital culture and skills play crucial amplifying roles, AI independently continues to exert a direct strategic influence—underscoring the multidimensional mechanisms through which digital technologies shape organizational competitiveness.

5. Discussion

The primary objective of this study was to examine the impact of AI on the sustainability of competitive advantage in the tourism and hospitality sector, with particular emphasis on the mediating roles of digital culture and digital skills. The findings of this study offer important theoretical and practical insights into the complex relationship between AI, digital enablers, and sustainable competitive advantage within the tourism and hospitality sector. The study's finding that AI has a significant and positive influence on sustainable competitive advantage reinforces the strategic value of AI when it is integrated holistically into the core operations of tourism and hospitality organizations, rather than being confined to isolated functions. This result aligns with prior research. For instance, Ref. [24] demonstrated that AI contributes meaningfully to achieving sustainable competitive ad-

vantage. Similarly, Ref. [25] emphasized the role of AI integration as a strategic enabler that supports long-term organizational competitiveness and sustainability. Reference [26] also identified AI as a critical strategic resource that enhances an organization's ability to maintain competitive positioning in dynamic and rapidly evolving market environments.

Equally noteworthy is the study's confirmation that AI has a statistically significant and positive impact on digital culture. This result is consistent with a growing body of literature. For example, Ref. [14] found that the adoption of AI significantly contributes to the development and reinforcement of digital culture in organizational settings. Similarly, Ref. [30] reported that AI integration promotes a digitally oriented organizational mindset, thereby strengthening the cultural foundation necessary for digital transformation. In the same vein, Ref. [32] emphasized that AI acts as a transformative force, embedding digital fluency and adaptability into everyday organizational practices, thereby reinforcing and institutionalizing digital culture.

Furthermore, the study reveals that digital culture exerts a significant and positive influence on sustainable competitive advantage. This finding aligns with prior research emphasizing the strategic importance of cultivating a digitally oriented organizational environment. For instance, Ref. [37] concluded that a strong digital culture significantly enhances an organization's capacity to build and sustain long-term competitive advantage. Similarly, Ref. [38] found that digital culture supports the development of sustainable competitiveness by fostering continuous innovation and strategic adaptability. Additionally, Ref. [39] argued that nurturing a digital culture strengthens strategic flexibility and innovation capacity—both of which are essential for sustaining competitive advantage over time.

In a parallel pathway, the study also found that AI positively influences digital skills, indicating that increased exposure to intelligent technologies encourages employees to acquire and enhance their digital competencies. This finding is consistent with several prior studies. Reference [54] reported that AI integration significantly boosts employees' digital capabilities by fostering continuous learning, facilitating technology adoption, and improving efficiency in digital tasks. Similarly, Ref. [55] concluded that AI contributes to the advancement of digital skills by reshaping job requirements and motivating employees to acquire new technological competencies. From a tourism and hospitality perspective, recent evidence reinforces this argument. For example, the deployment of AI-powered concierge robots and self-service kiosks in international hotel chains has required frontline staff to develop new competencies in managing, troubleshooting, and coordinating with intelligent service systems [1,9,83]. Likewise, the integration of AI-based recommendation engines and chatbots in travel agencies has compelled employees to acquire skills in digital content management and customer data analytics. Empirical studies in the hotel sector further indicate that staff exposed to smart property management systems and AI-driven booking platforms report higher proficiency in digital problem-solving and system navigation [83–86]. These cases illustrate that AI adoption in tourism and hospitality does not replace employees but rather stimulates the upgrading of their digital skills, aligning with the broader evidence on skill development in technology-intensive service industries.

The significant positive effect of digital skills on sustainable competitive advantage reinforces the strategic importance of digitally empowered human capital as a key differentiator in competitive environments. This finding aligns with previous research. Reference [62] concluded that digital skills positively influence the achievement of enduring competitive advantage. Similarly, Ref. [63] found that digital capabilities significantly contribute to competitiveness, particularly in technologically volatile contexts, by enhancing adaptability, innovation, and digital responsiveness. Ref. [64] argued that strengthening digital competencies across the tourism workforce is essential for achieving long-term competitive

advantage, particularly when complemented by green skills that promote adaptability and innovation in a rapidly evolving environment.

Crucially, the study reveals that digital culture partially mediates the relationship between AI and sustainable competitive advantage. This indicates that while AI exerts a direct influence, its full strategic benefits are only partially realized through the cultural context—specifically the shared values, norms, and behaviors that characterize the organizational environment. Similarly, the finding that digital skills serve as a partial mediator between AI and competitive advantage highlights that the human element does not completely determine outcomes but plays a crucial role in amplifying and channeling the value of AI toward sustainable advantage. Taken together, these findings suggest a dynamic interrelationship between digital culture and digital skills. A supportive digital culture provides the norms and openness necessary for continuous learning and experimentation, thereby fostering the development of digital skills across the workforce. In turn, employees equipped with advanced digital skills reinforce and sustain this culture by translating shared values into tangible practices, innovations, and routines. Without a conducive culture, digital skills may remain underutilized; without adequate skills, cultural values may not materialize in actual digital transformation. Hence, the two enablers operate synergistically, and their interplay forms a reinforcing cycle through which AI adoption is embedded more deeply, ultimately securing sustainable competitive advantage [19,27,31,36].

6. Theoretical Implications

This study makes a substantive theoretical contribution by extending the TAM beyond its traditional focus on individual user behavior and applying it to an organizational context shaped by digital transformation and strategic performance outcomes. While TAM has typically emphasized perceived usefulness and perceived ease of use as determinants of technology acceptance, the findings of this research show that these perceptions are embedded within, and shaped by, the broader organizational environment—particularly through digital culture and digital skills.

A central result is that AI significantly influences both digital culture and digital skills, suggesting that perceptions of AI are not formed in isolation but emerge from socially constructed, interactive experiences within the workplace. This challenges the linear, individual-level assumptions of classical TAM and offers a more contextualized view: acceptance is not simply a cognitive judgment but a process shaped by shared values, collective learning, and organizational competencies. In this way, digital culture and digital skills function not just as mediators but as amplifiers that enhance and sustain the behavioral outcomes TAM seeks to explain.

Furthermore, the evidence that digital culture and digital skills each exert direct effects on sustainable competitive advantage adds a strategic dimension rarely explored in TAM-based studies. It shows that the implications of technology acceptance go beyond initial adoption, positioning TAM as a useful framework for understanding how acceptance contributes to long-term organizational capabilities and competitive positioning when embedded in digitally maturing contexts. Coupling TAM with constructs of readiness and adaptability enhances its explanatory power, revealing deeper connections between perception, behavior, and performance.

Finally, applying TAM to AI adoption in Saudi Arabia's tourism and hospitality industry—an innovation-intensive sector undergoing rapid transformation—confirms the model's flexibility and relevance in emerging market settings. The study demonstrates that TAM can be meaningfully adapted to explain not only consumer or frontline adoption but also the organizational dynamics of enterprise-level technological integration. This positions TAM as a living model—capable of evolving in scope and sophistication when

applied to complex, real-world challenges such as sustaining competitive advantage in the digital era.

Figure 2 illustrates how the study integrates its theoretical framework with the methodological approach. Artificial intelligence (AI) serves as the primary independent construct, shaping both TAM perceptions (perceived usefulness and perceived ease of use) and the organizational enablers of digital culture and digital skills. These factors, in turn, exert direct and mediating effects on sustainable competitive advantage. Structural Equation Modeling (SEM) provides the methodological mechanism through which these hypothesized paths are empirically tested, thereby linking the conceptual framework with the analytical strategy.

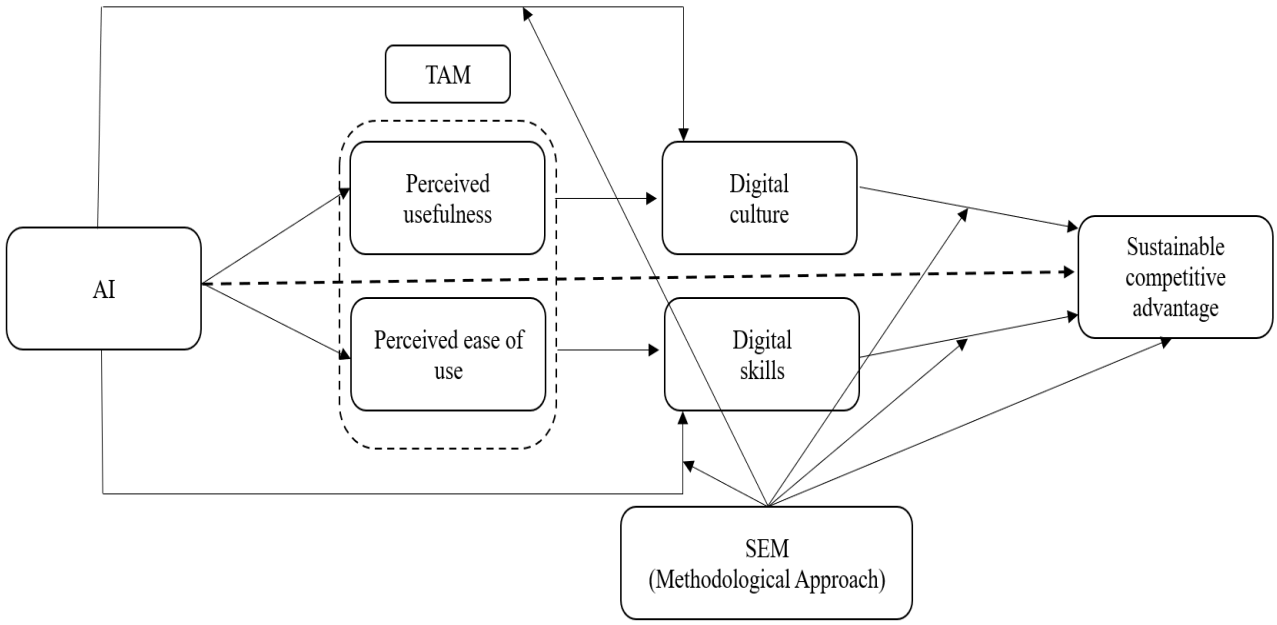

Figure 2. Process Diagram Linking Theoretical Framework (TAM) and Methodology (SEM).

7. Practical Implications

To unlock the strategic potential of AI and translate it into sustained competitive advantage, tourism and hospitality organizations in Saudi Arabia need to adopt a holistic approach that extends well beyond the acquisition of new systems. The process begins with aligning AI technologies to clear operational and service priorities where they can deliver measurable value—such as guest personalization, predictive analytics, and dynamic resource management. Yet the effectiveness of such integration depends on the organization's internal readiness. This makes it essential to conduct capability audits before adopting AI, ensuring that the chosen systems are appropriate for the current level of digital maturity and can be scaled gradually as capabilities evolve.

Equally important is the deliberate cultivation of a digital culture that encourages continuous learning, experimentation, and openness to change. Such a culture cannot be left to develop informally; rather, it must be embedded intentionally into performance frameworks, internal communications, and leadership practices. Senior managers and department heads play a central role by acting as digital role models—actively using and promoting AI tools in daily operations—while recognition and reward systems should highlight teams that embrace digital innovations or contribute to AI-driven improvements. Embedding these behaviors across all levels of the organization helps to build collective confidence in technology and reinforces the shared norms that are vital for long-term cultural transformation.

Workforce development is another critical element. Training must move beyond one-off sessions toward targeted programs that are closely tied to real organizational tasks and systems. Tourism and hospitality organizations should develop ongoing learning ecosystems—such as AI learning labs, cross-departmental projects, and micro-certification programs—where employees can practice applying AI in business-relevant contexts. These initiatives should be tailored to different organizational roles, ensuring that frontline staff, supervisors, and executives alike develop the digital capabilities most relevant to their responsibilities. The strong effect of digital skills on sustainable competitive advantage, as confirmed by this study, underscores the importance of building competencies that extend beyond technical proficiency to include adaptability, interpretation of AI outputs, and system optimization.

The mediating roles of digital culture and digital skills further highlight the need to integrate technology strategies with organizational development frameworks. AI cannot generate long-term benefits unless accompanied by deliberate efforts to close the cultural and capability gaps that often hinder its impact. Managers should implement mechanisms to monitor digital engagement, foster interdepartmental collaboration, and track cultural alignment as part of AI readiness assessments. Where gaps are identified, proactive measures—such as leadership coaching, digital mentoring, and internal campaigns to normalize technology use—should be undertaken.

Finally, at the policy level, sector leaders guiding Saudi Arabia's Vision 2030 transformation—such as the Ministry of Tourism, the Human Capability Development Program, and digital transformation units—must recognize that sustainable competitiveness in the digital era requires investment in both technology and people. National initiatives that encourage AI adoption should be paired with structured support for organizational change management, digital leadership development, and sector-wide digital literacy programs. Establishing benchmarks for digital culture and skills, combined with recognition mechanisms for digital excellence, would help accelerate the spread of best practices across the industry. In this way, AI will not remain an isolated technical tool but will evolve into a shared capability embedded in the very fabric of Saudi Arabia's tourism and hospitality sector.

8. Limitations and Future Research

While this study provides valuable insights into the strategic role of AI, digital culture, and digital skills in the Saudi tourism and hospitality sector, certain methodological and contextual limitations must be acknowledged. Recognizing these constraints not only clarifies the study's scope but also opens avenues for future research to deepen and expand upon its findings. First, the study was conducted exclusively in Saudi Arabia, a country undergoing rapid, top-down digital transformation within the framework of Vision 2030. While this context adds strategic relevance, the findings may reflect cultural, institutional, or regulatory dynamics specific to the Kingdom. Future studies are encouraged to conduct cross-country comparative research, particularly within the GCC or broader MENA region, to explore how national context moderates the relationship between AI, digital enablers, and competitive outcomes in the tourism and hospitality sector.

Second, the study's sample was limited to managers and supervisors with $\geq$1 year of tenure working in five-star hotels, travel agencies, and destination management companies. Although this focus ensures informed managerial perspectives, it narrows generalizability by excluding SMEs and frontline employees. Moreover, because purposive sampling was employed and exact counts of managers and supervisors per stratum were not available, representativeness is limited. Future research should adopt probability-based or multi-level

sampling designs, supported by administrative workforce data, to improve generalizability and reduce sampling bias.

Third, although the study highlights the importance of digital culture and digital skills as mediating variables, it treats these constructs as single-layer, aggregate factors. In reality, both culture and skills are multi-dimensional and may exhibit internal variation across departments, roles, or units. Future research should adopt a multi-dimensional analytical approach to unpack the specific components of digital culture (e.g., innovation orientation, knowledge sharing) and digital skills (e.g., data literacy, AI interaction competence) and investigate which sub-dimensions most strongly mediate the effect of AI on organizational performance.

Fourth, the current study focused on the impact of AI on sustainable competitive advantage through the mediating roles of digital culture and digital skills. Future research, however, should expand this scope by examining additional determinants of sustainable competitive advantage such as service innovation, customer engagement, and dynamic capabilities. Moreover, given the rapid technological transformations reshaping the tourism and hospitality sector, further studies could explore AI-related variables including trust in AI systems, ethical use of AI, and employee adaptability, thereby opening new horizons for understanding the diverse ways in which AI influences organizational performance and workforce experiences.

Fifth, the constructs of digital culture and sustainable competitive advantage were measured using concise, previously validated scales. While this approach ensured reliability and comparability, it inevitably simplifies complex, multidimensional concepts. Future studies should expand measurement by incorporating sub-dimensions—such as innovation orientation and knowledge sharing for digital culture, or dynamic capabilities and market responsiveness for sustainable competitive advantage—in order to capture these constructs more comprehensively.

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Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A Population and Sample Distribution by Subsector and Region (N = 488)

Table A1. Population figures are based on official Ministry of Tourism records (2025). The sample size reflects managers and supervisors with ≥1 year of tenure; more than one respondent could be surveyed from the same organization.

Region	5★ Hotels (Institutions)	5★ Hotels (Respondents)	Travel Agencies (Institutions)	Travel Agencies (Respondents)	DMCs (Institutions)	DMCs (Respondents)
Riyadh	34	60	68	59	25	35
Jeddah	20	35	48	41	26	36
Makkah	21	37	34	29	13	18
Madinah	20	36	17	15	15	21
Eastern Province	16	28	39	34	3	4
Total	111	196	206	178	82	114

Note: Population figures reflect the number of licensed institutions [72]. The sample size exceeds the number of institutions because multiple eligible managers and supervisors were surveyed per organization.

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